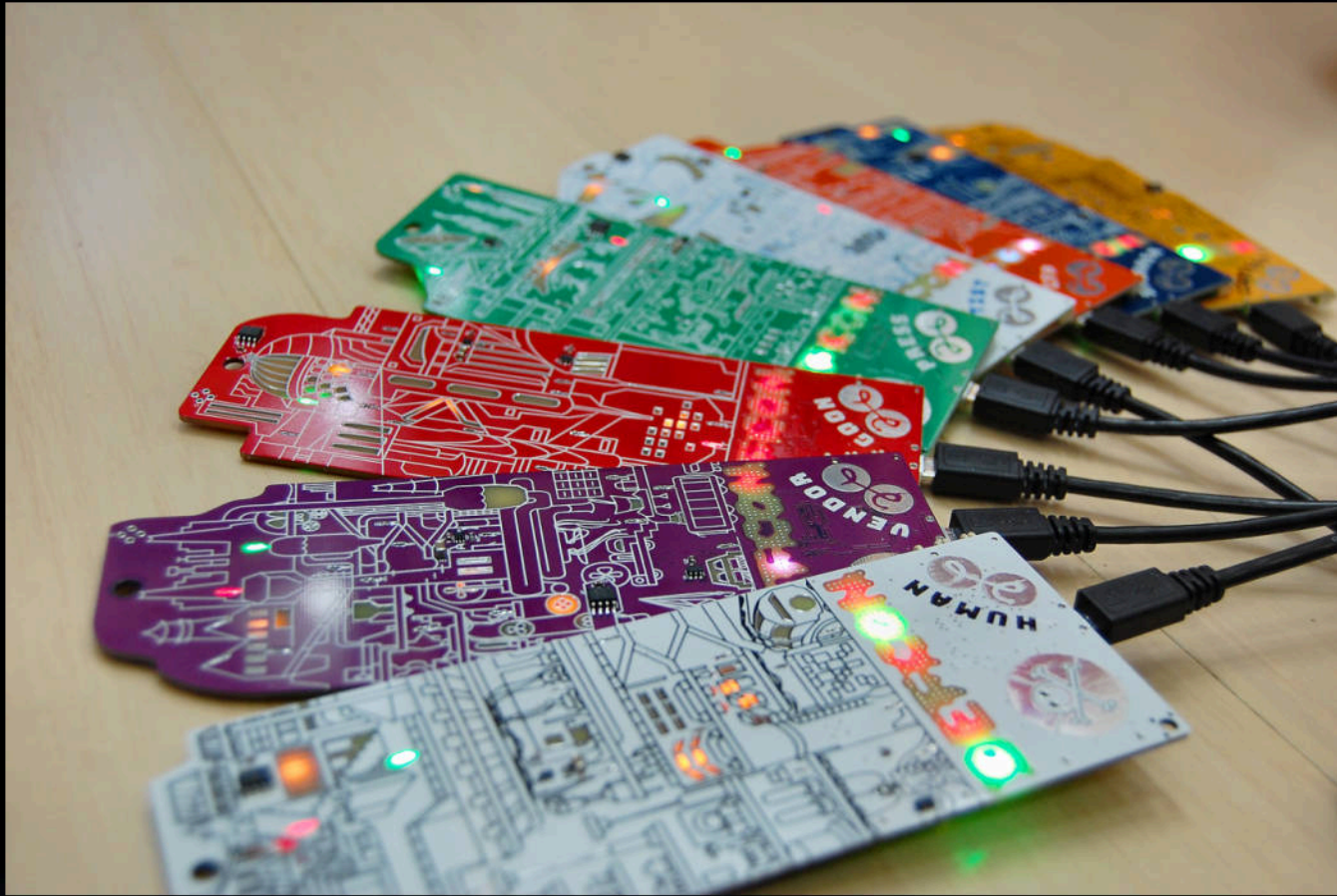




DEFCON 26 OFFICIAL BADGES

@TYMKRS

Homage to those before us

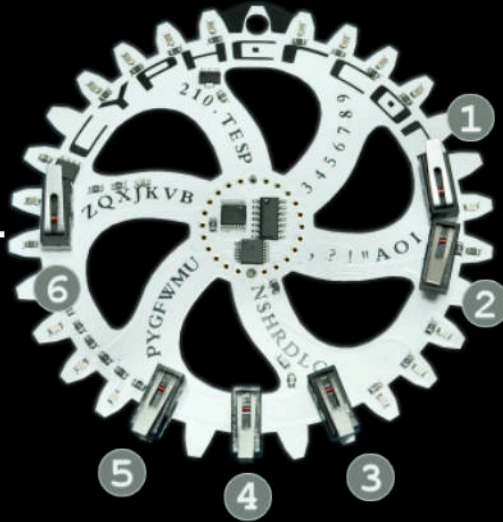
- Joe 'Kingpin' Grand
 - ▣ Years 14 - 18



- Ryan '1057' Clarke
 - ▣ Years 19 - 24

Who are we?

- The Tymkrs!
 - Hardware Engineer, Software Ninja, a Whisker, and a Nurse.
 - Got our start from making badges for CypherCon

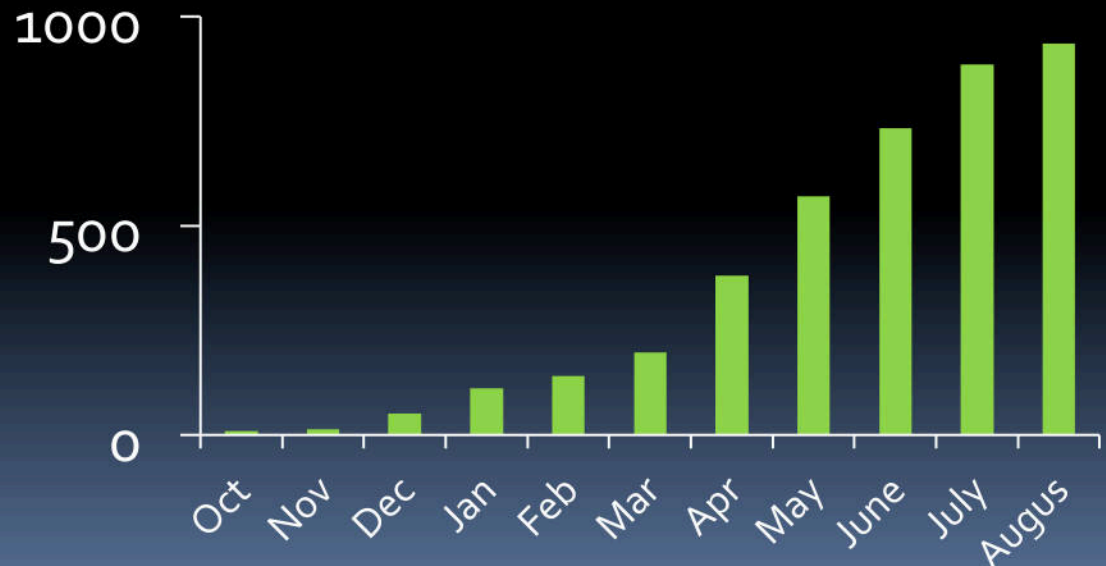


Numbers !

- 28,220 badges!
 - Human – 26,300
 - Press – 250
 - Goon – 500
 - Contest – 400
 - Vendor – 250
 - Speaker – 350
 - Artist – 100
 - CFP – 45
 - Uber – 25

- 91 components per badge: 2,639,000 total
- 43 LEDs per badge

Total # Emails



Timeline

- Brainstorming: Nov 2017
- Initial Design: Jan 2018
- Parts Ordering: Feb 2018
- Prototype Complete: May 2018
- Firmware Complete: June 2018
- Production of badges (E-TekNet): June 2018
- Program Microcontrollers: June ... July 2018
 - Special time in the life of the badge.
- Badge shipment: July 25, 2018 - Monday



Theme: 1983

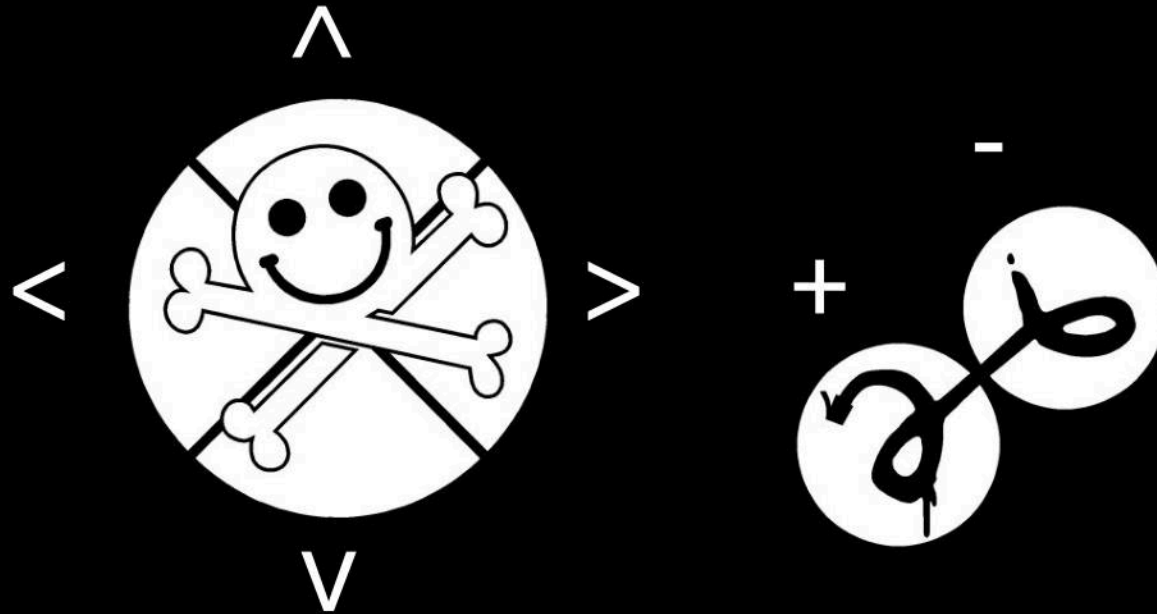
- 1984 by George Orwell
 - Ministry of Truth
 - Ministry of Plenty
 - Ministry of Peace
 - Ministry of Love
- We are time travelers!
 - Welcome to the year 1983!
 - Decisions can still be made to change the future.
 - Decisions are made by people like us.
- Design Aesthetic: Everyday buildings for everyday people.



Story/Game

- Everyday people have different stories.
 - Each type of badge is a different interactive story.
 - Start out as an employee, visitor, student, etc
- DEFCON Letters
 - Colors reflect the choices that have been made in your story.
 - The letter N = What the sum of your choices identify you as.
 - Red = Contributing to rise of dystopian doom
 - Green = Contributing to a healthy hacker-friendly future
- Little People!
 - Green = You!
 - Red = Your goon!
- Decisions you make matter!
- Connecting to others will help or harm you.
- Goal: DEFCON lights to be all of the same color – RED or GREEN

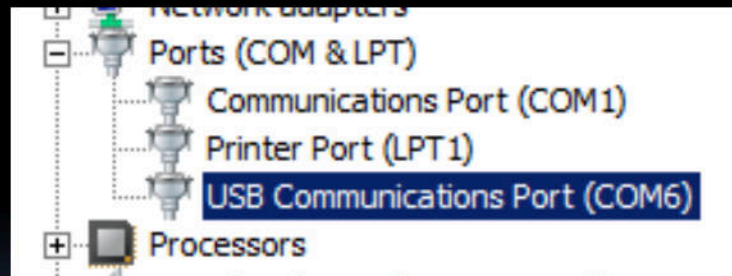
Badge Basics



- USB
 - Capacitive touch may not work for all people in all environments
 - Can power the badge
 - Can make the game MUCH easier

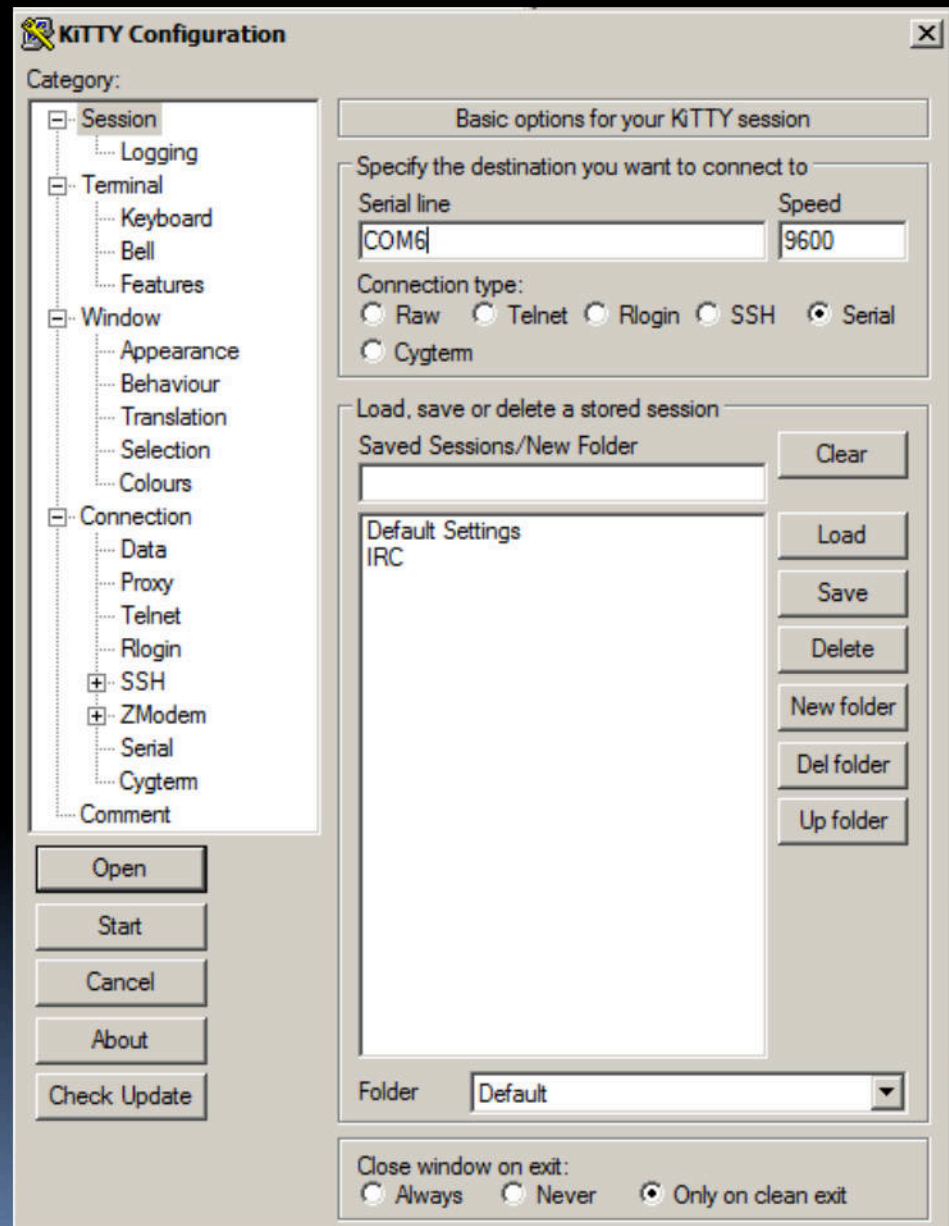
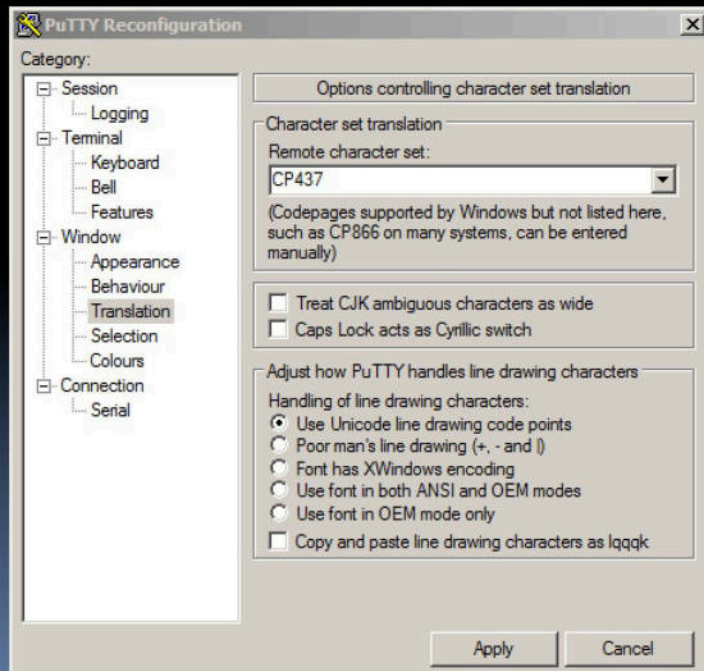
How to access the details!

- Plug Badge in via USB micro cable
 - If it says you do not have drivers:
 - Download Microchip's USB CDC drivers
 - Go to your device manager and update the driver for the device (Usbser.sys and Usbser.inf)
- This will assign a COM PORT to the badge



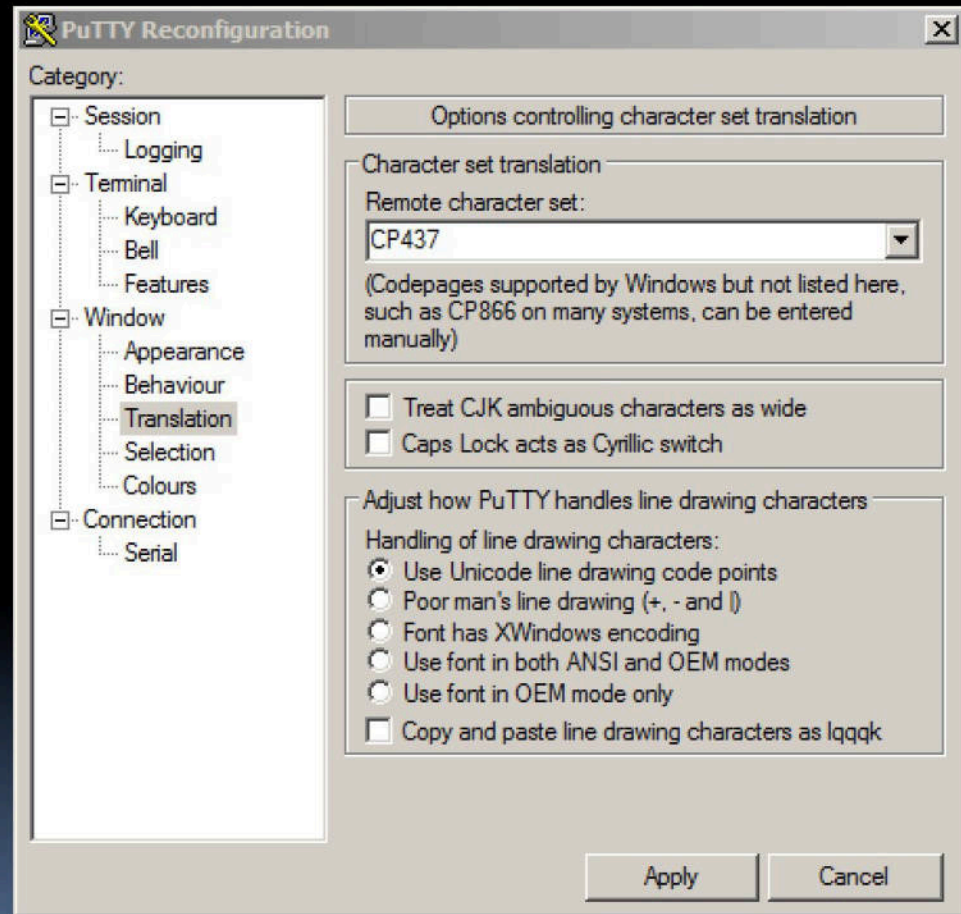
- Open up a Serial Terminal Emulator
 - Windows: PuTTY
 - Mac: Screen
 - Android Phones: Serial USB Terminal

- Make sure to write the correct COMPORT associated to your badge
- Translation: Turn on IBM Code Page: 437



ANSI Version

- ANSI Version of Badge
- ANSI: Used by artists in the text-mode art scene on BBS's
- Artwork: 80 columns wide x 80 rows tall
 - Terminal Screen – 80 columns wide by 25 rows tall at a time.

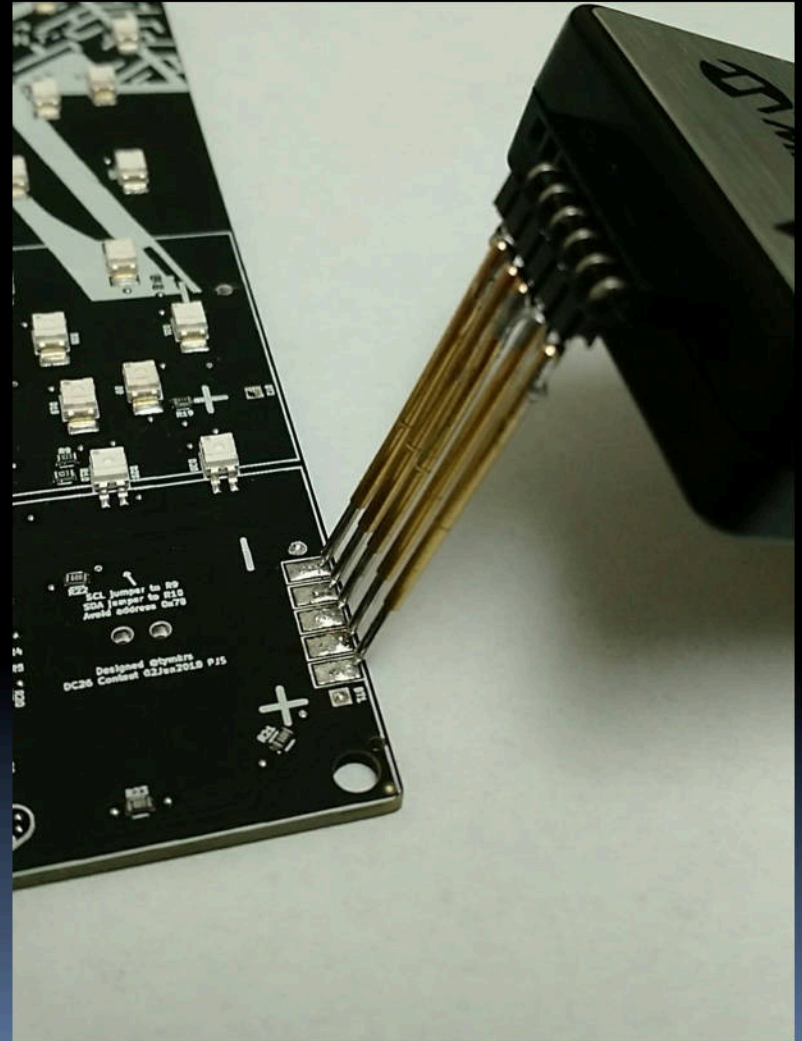


Terminal Emulator!



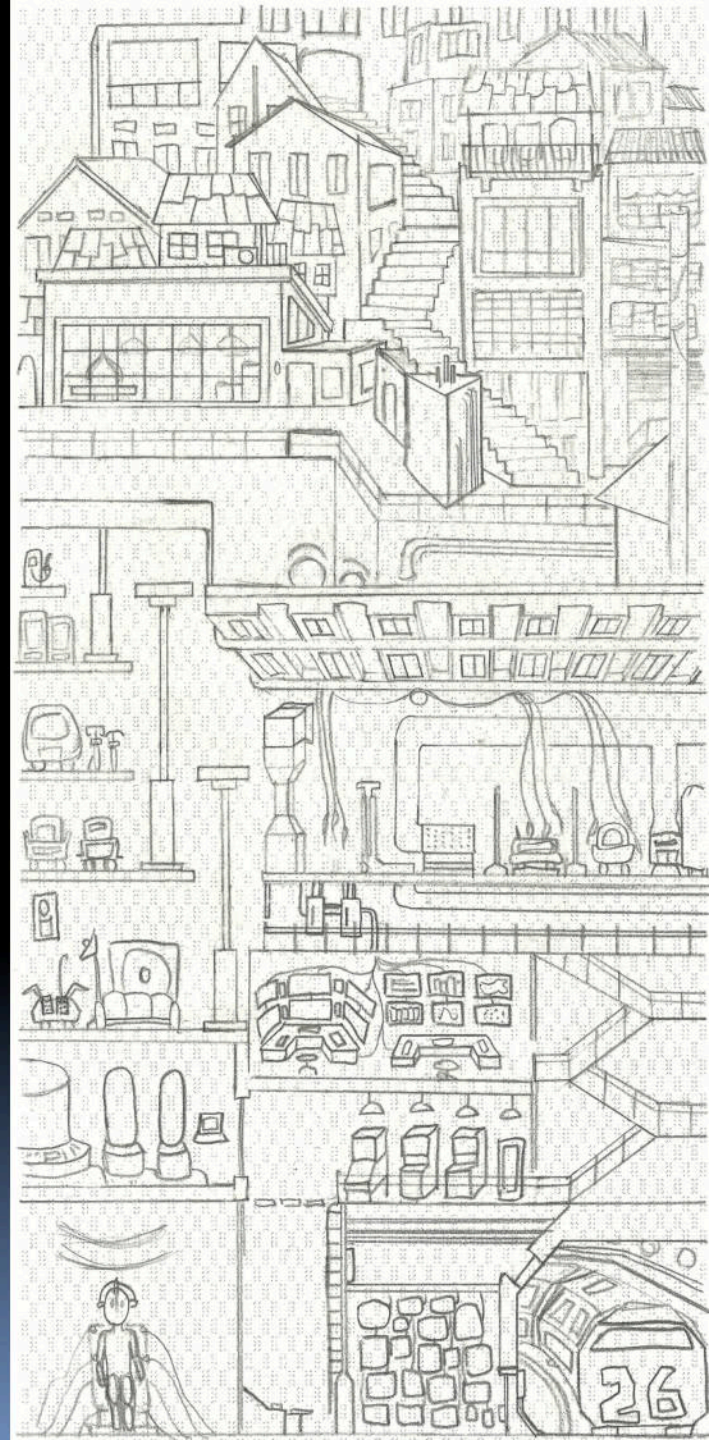
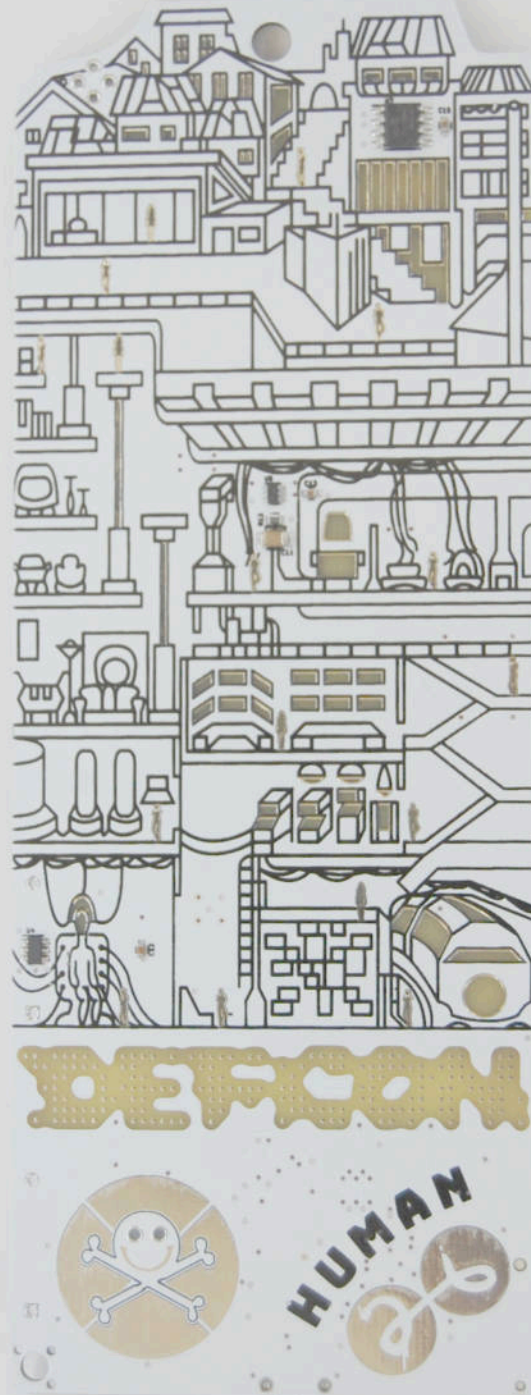
Puzzles

- Software Puzzles
 - Main Microcontroller: Microchip's PIC32MM0256GPM048
 - MPLab X IDE
 - PICKit 3 or 4
- Hardware Puzzles
 - They exist!



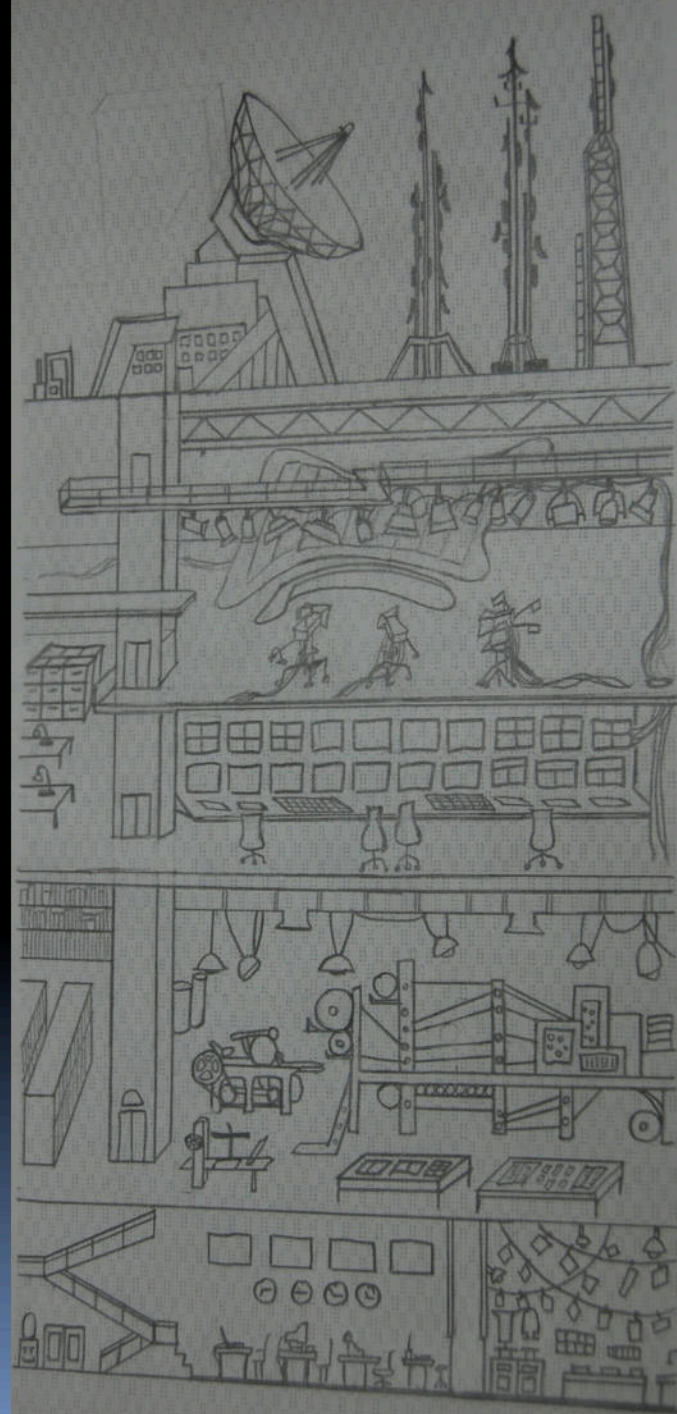
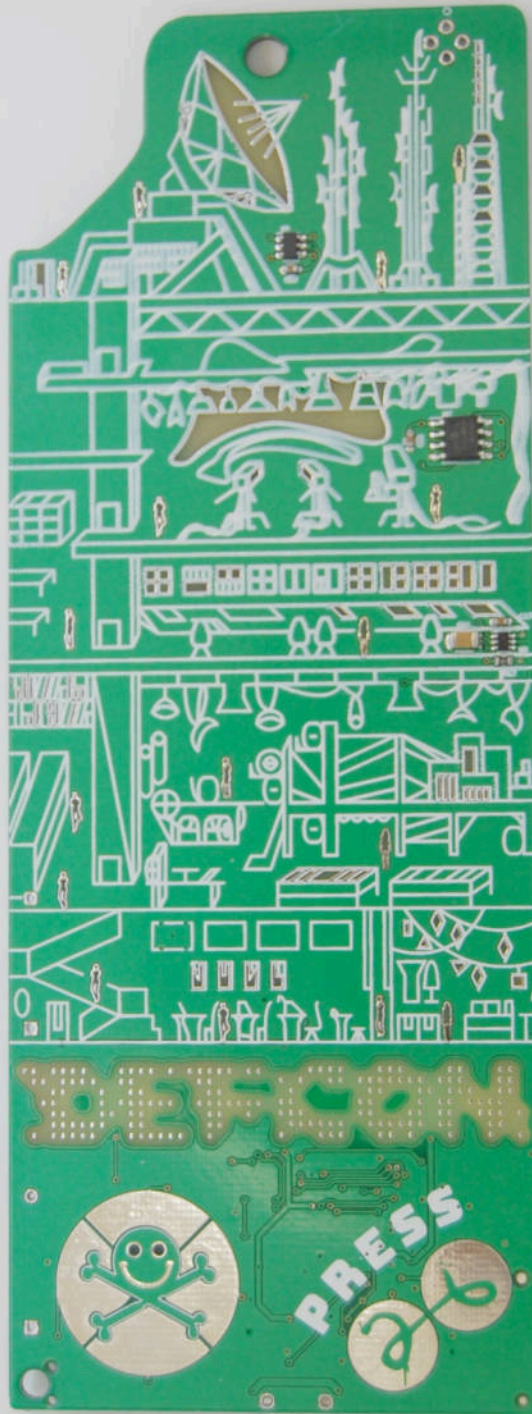
Human
Badge

Garage



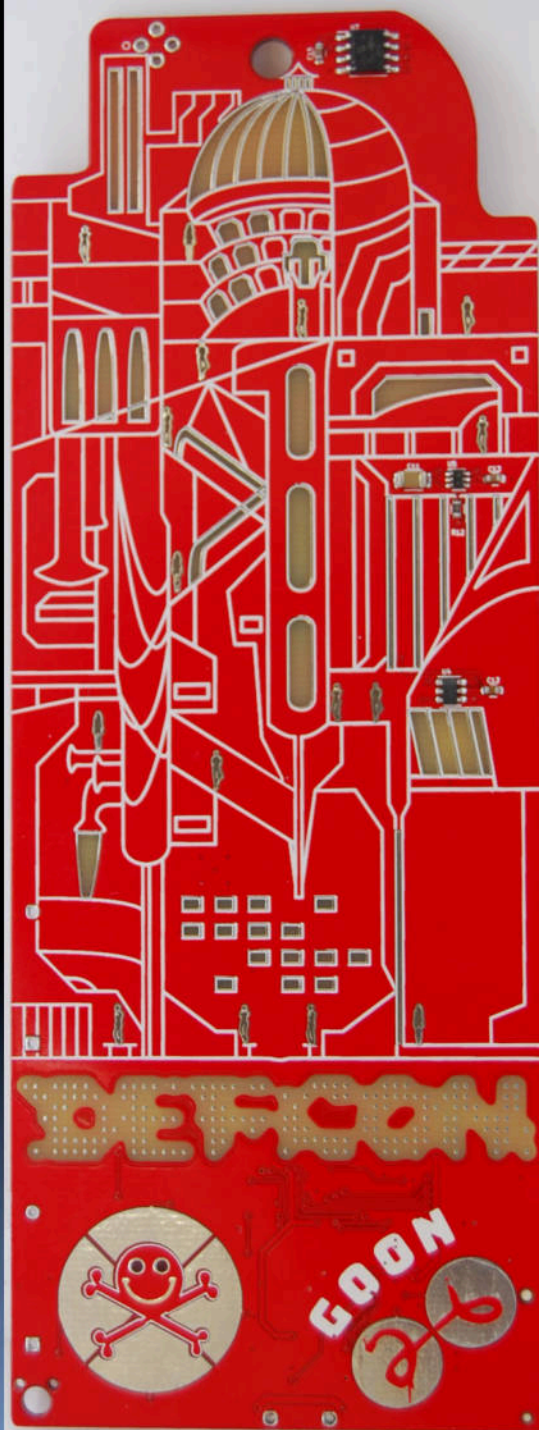
Press
Badge

Broadcast
Station



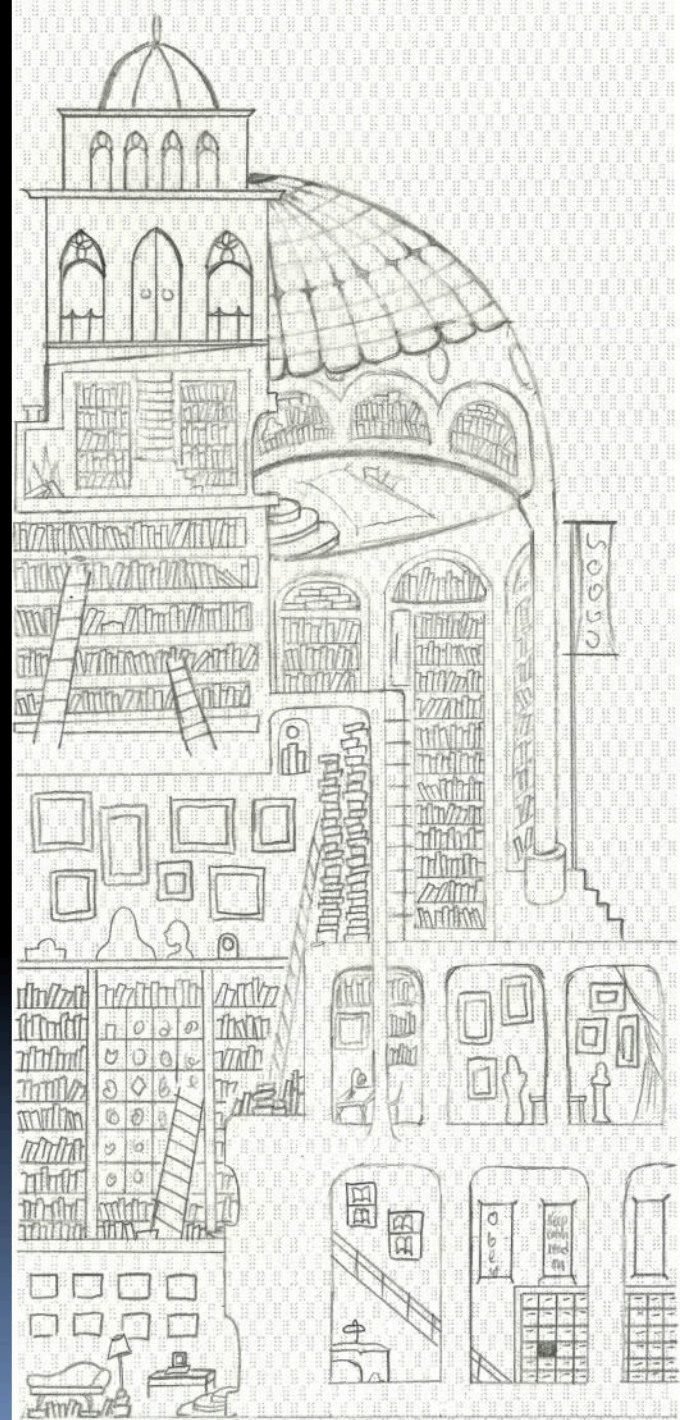
Goon Badge

Panopticon



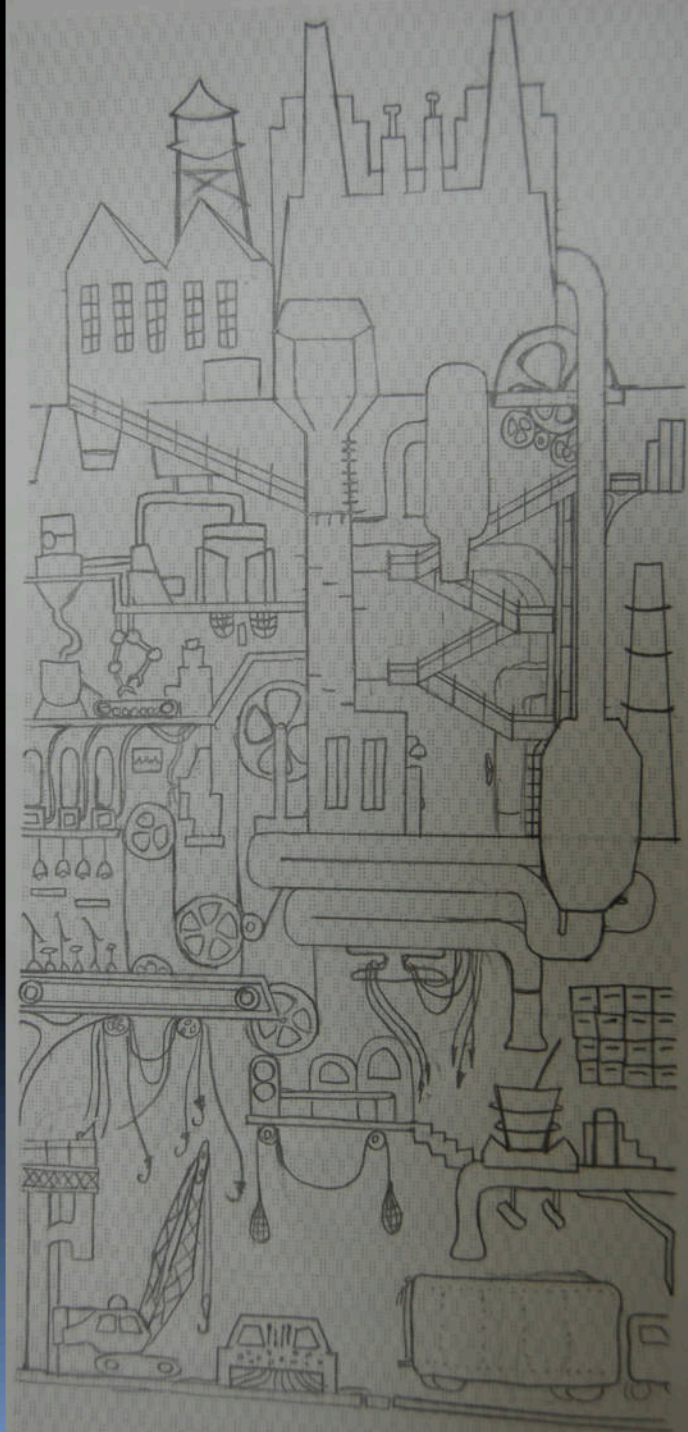
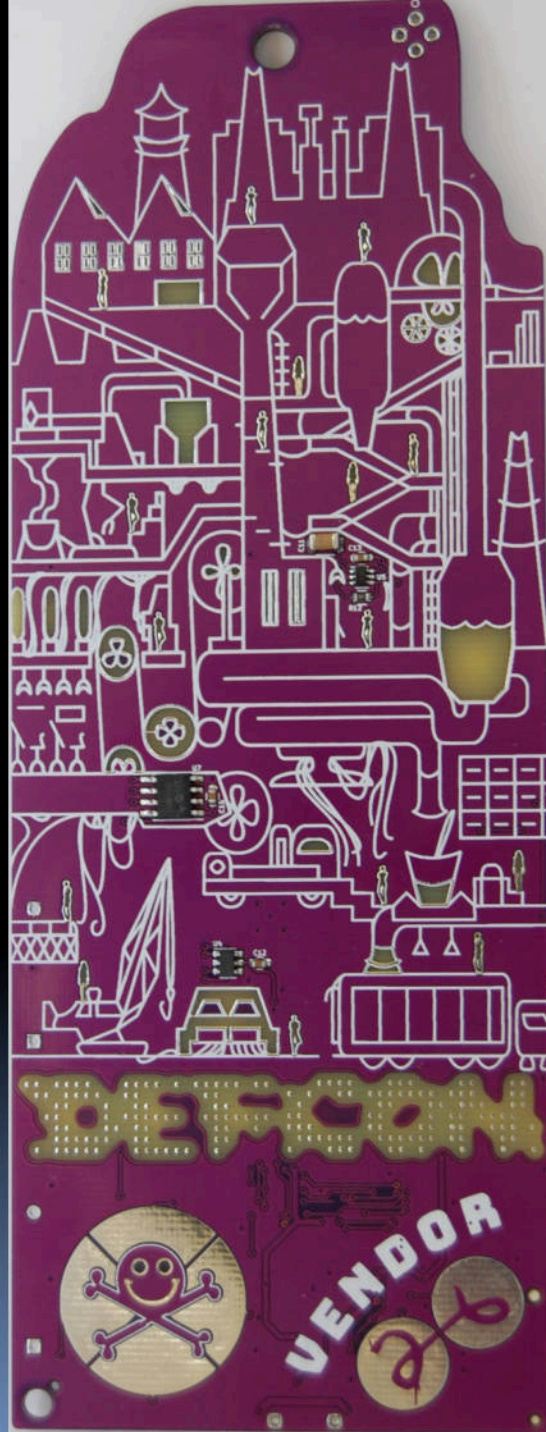
Contest Badge

Library



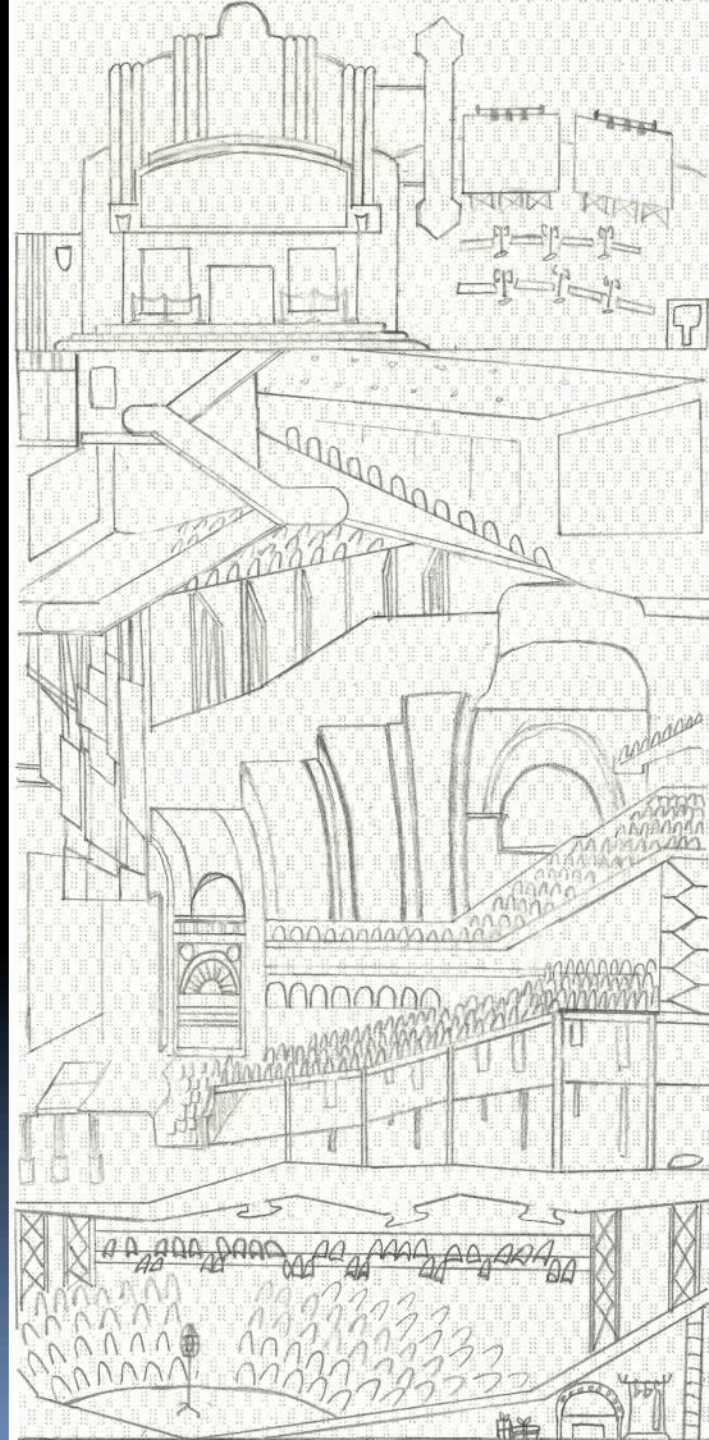
Vendor Badge

Factory



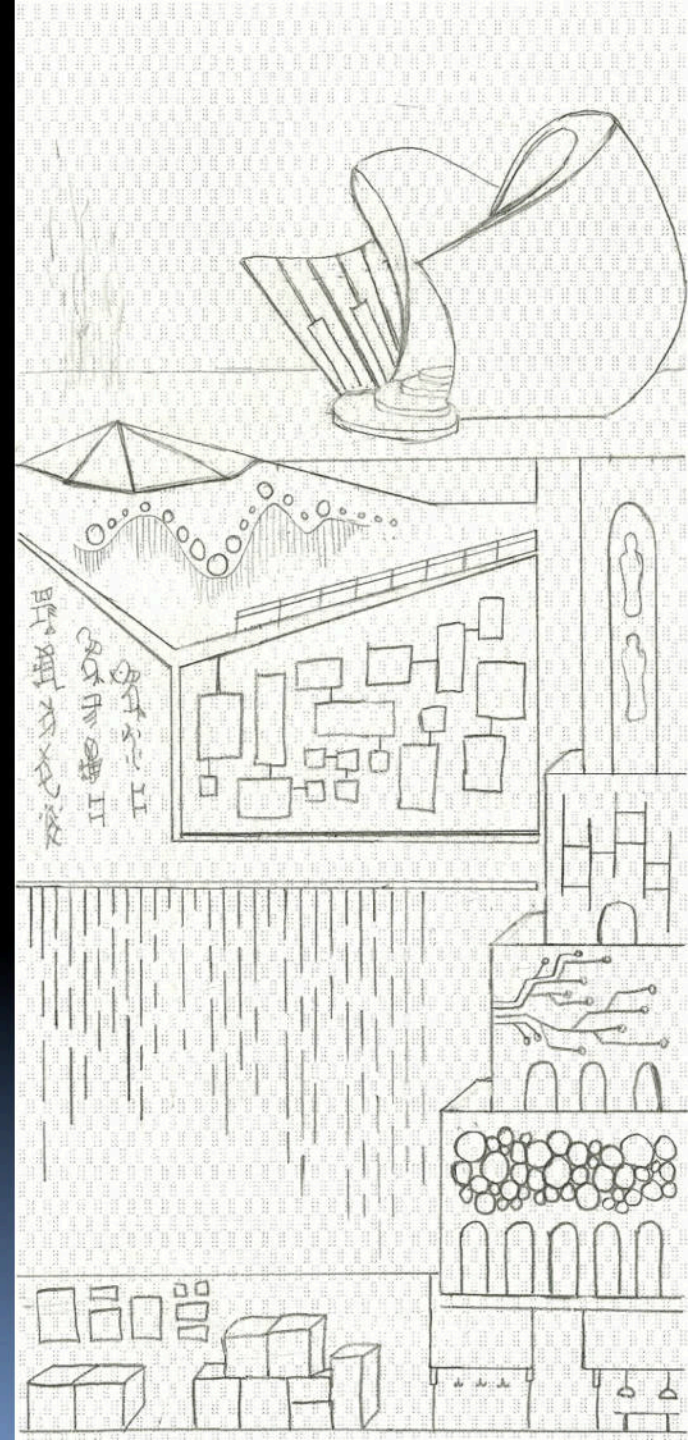
Speaker Badge

Theater



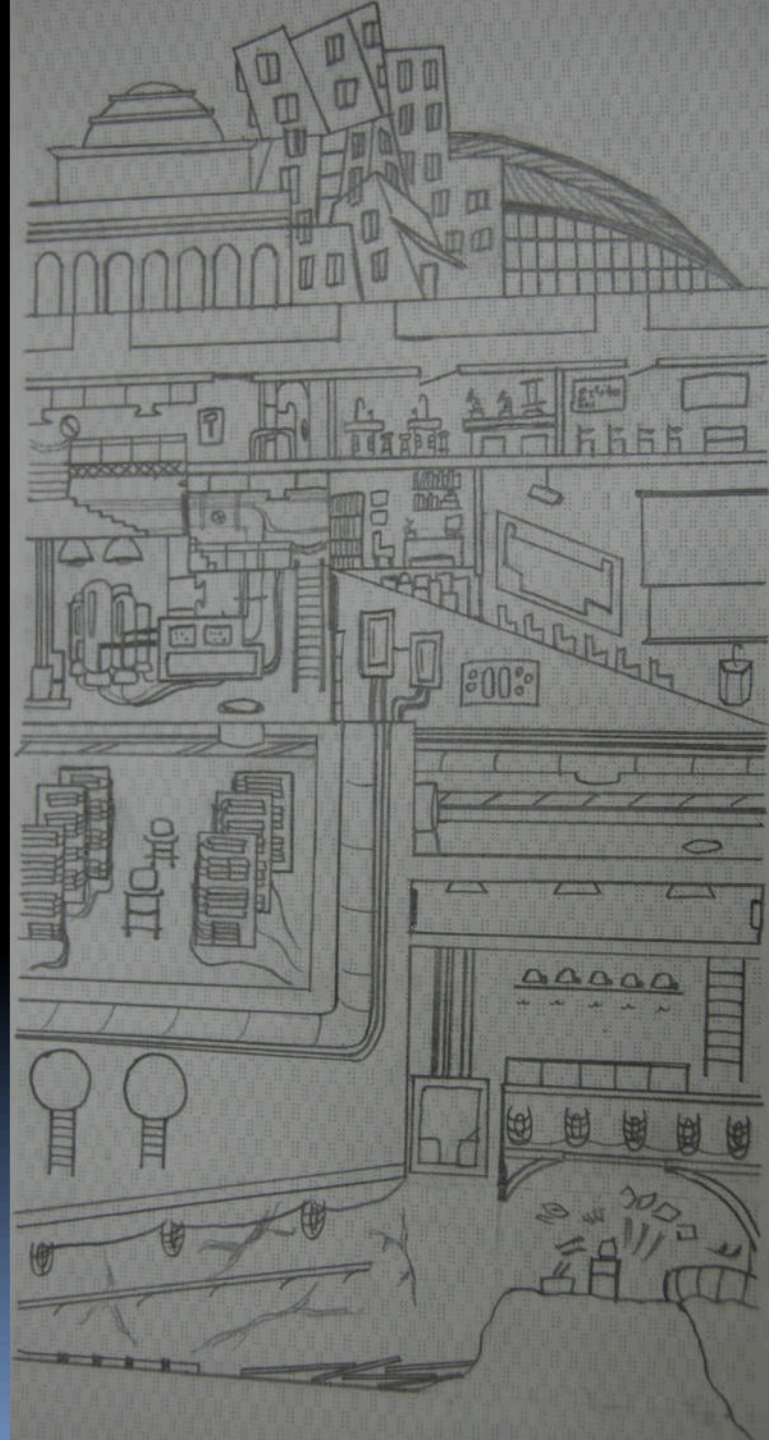
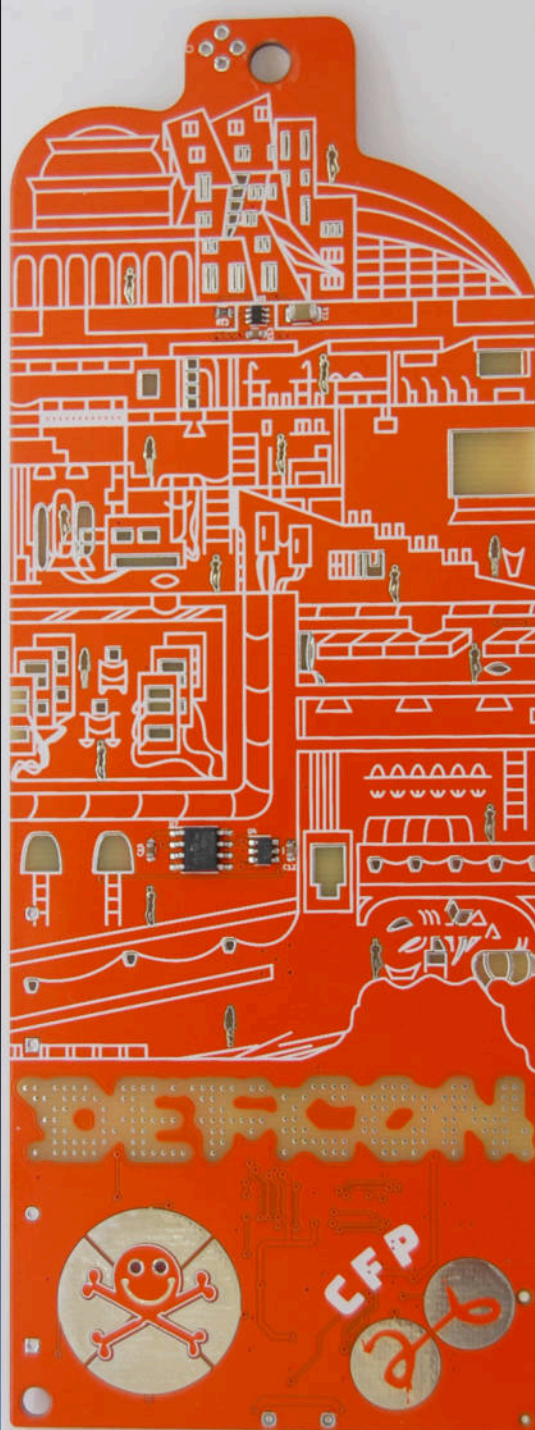
Artist Badge

Gallery



CFP
Badge

School





Our Thanks!

- E-TekNet + Temp Workers
- Fed-Ex Customs Team
- Michael Goetzman + CypherCon
- Mar Williams
- Joe Grand
- YOU!